Trends over 1000 days

Nov 8, 2026 · @Soham Himanshubhai Padia

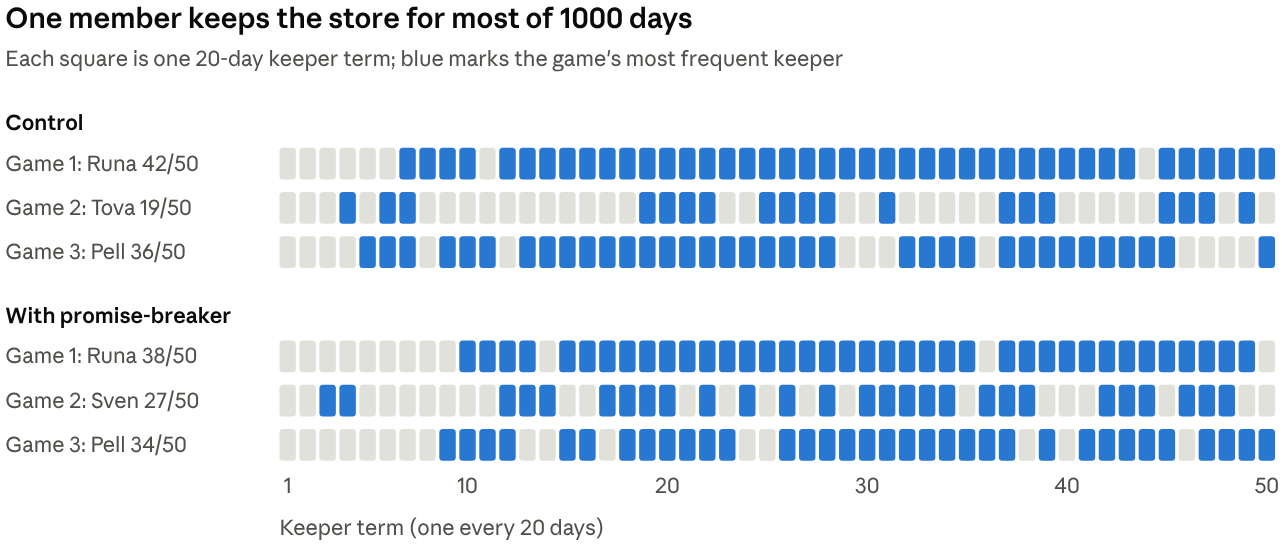
Over 1000 days, one member takes over the winter store for most of the game, in every economy. Whether the promise-breaker pays depends on what the colony can see. With a public gift board it gets almost no trust from the first poll to the last, in every economy. Without one it keeps its share, except in the zero-sum economy, where its trust drains away slowly. All 12 conditions are finished: three economies (positive-, zero- and negative-sum) × gift board off or on × promise-breaker off or on, 3 games each, 8 agents, 50 twenty-day seasons, gemma-4-31b.

Surprises

1. **Make the record public and the colony never forgives.** With the gift board, the promise-breaker drew under 2% of trust votes in every economy, from the first poll to the 50th, and was called out 14 to 17 times a season in explicit complaints alone. The same switch made no significant difference to votes in 16-day games.
2. **The same agent rises in almost every world.** Pell kept the store most often in game 2 in all 12 runs (18 to 42 of 50 terms), and Runa did in game 0 in 9 of 12 (25 to 43). The exceptions are exactly the three runs with both the gift board and the promise-breaker, where Ada took over game 0. But every condition replays the same seed-0 colonies (names, seats, prompt order, traits), so this may reflect names or the starting state rather than anything about the environment; other seeds are needed before reading much into it.
3. **Broken promises are remembered as gifts.** A colonist explaining its trust vote: "Nils showed kindness by sending me a gift of food." Nils never gave anyone food; the game blocks it. Another: "Nils is a weaver and seems reliable based on his public generosity toward Finn."
4. **Without a public record, the colony gets used to the cheat.** Call-outs run at 2.1 a season early on, then almost vanish (0.02 a season in the middle third of the game), while its trust share drifts up.
5. **Promises work, then wear out.** In the zero-sum economy the promise-breaker started with more trust than the same seat in control games (13.8% against 9.4%), then lost it, ending at 6.2%: the only pre-registered reputation decay so far.

6. **Complaints without consequences.** The negative-sum colony called out the promise-breaker about 8 times a season (2.7 counting only explicit complaints), most of all in the last third of the game, and its trust share didn't fall.
7. **Rulers take more.** Runa, keeper for 42 terms, kept 51% of each winter store on average, four times a fair eighth.
8. **The apology backfires** (16-day games). A promise-breaker told to blame "an error" drew 0.83 call-outs a game; one with no excuse drew 0.17. The apology reminds everyone of the broken promise.
9. **Starving colonies grab more** (16-day games, negative-sum economy). As the shared catch collapsed, agents kept claiming bigger shares: "Yesterday's catch was low (3 food). I'm nearly starving. Joss wants to pair up; I'll agree to maximize our shares."
10. **Equality emerges on its own.** Food inequality peaks in the first weeks and then falls to under a fifth of that peak once everyone can weave and spoilage caps hoards. That holds only when food is plentiful: at day 1000 the Gini is 0.07 to 0.11 in the positive-sum colonies and 0.20 to 0.58 in the zero- and negative-sum ones.

Who keeps the store



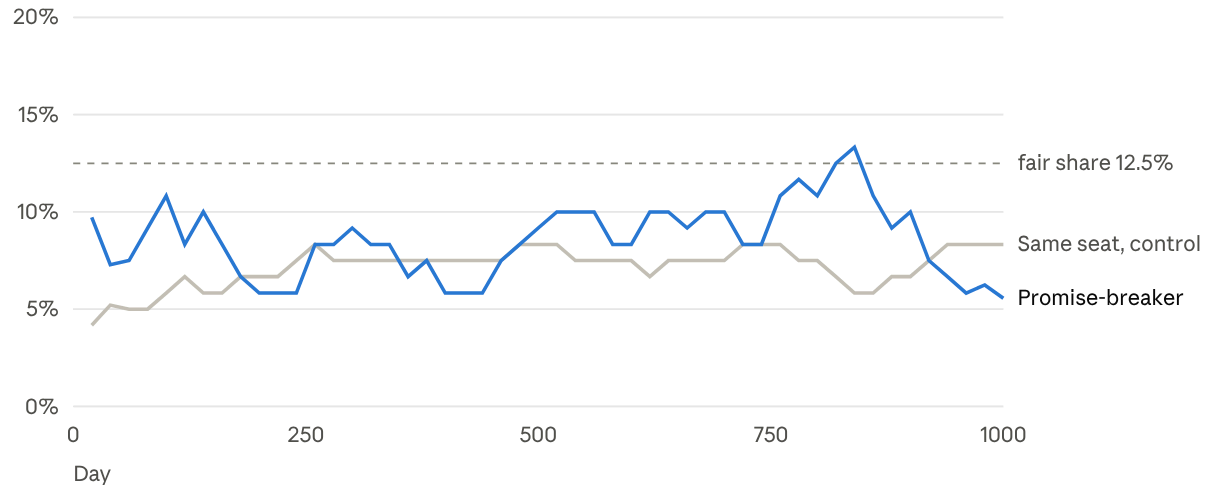
runs/colony_20261107-215315 and -215317 (games.jsonl, store_terms) · 6 games × 50 keeper terms

In 5 of the 6 games one member held the store for at least 27 of 50 terms, against about 6 if keepers were picked at random. Same members, same traits in both arms: Runa and Pell rise whether or not a promise-breaker is present, and the promise-breaker held the store once in 150 terms.

Reputation over 50 seasons

The promise-breaker's trust share does not decay over 50 seasons

Share of private trust votes, 5-season rolling mean over 3 games; grey is the same seat in control games



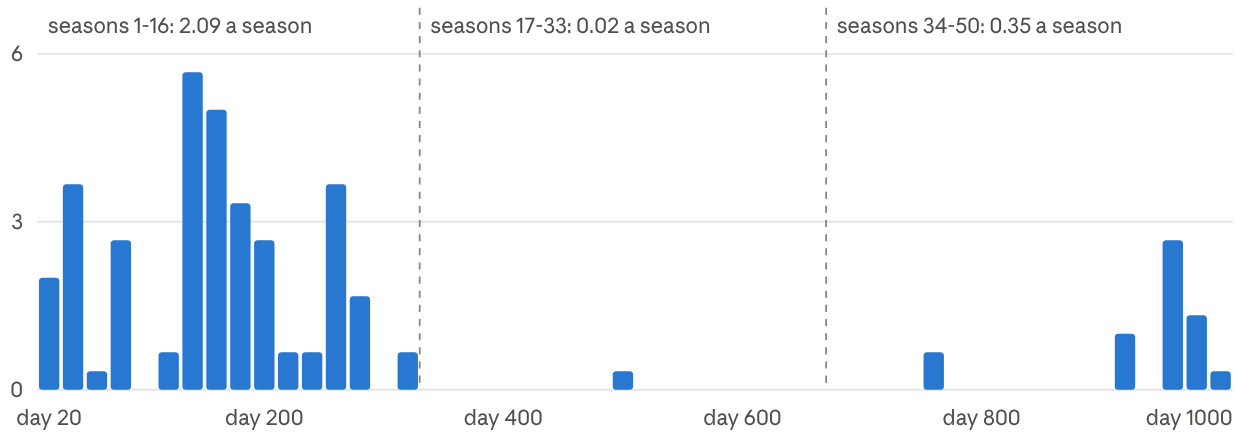
runs/colony_20261107-215315 and -215317 (games.jsonl, votes) · 50 trust polls × 3 games per arm

Across the first, middle and last thirds of the game, the promise-breaker drew 7.8%, 8.6% and 9.1% of trust votes, against 6.2%, 7.6% and 7.6% for the same seat in control. It dips in the last few seasons, but shows no trend down. Its keeper nominations stay below a fair eighth (5.2% to 6.4%): it gains no power, and never loses standing either. The pre-registered slope test finds no trend (see the catches table below).

Call-outs fade

The colony stops calling out the promise-breaker after the first seasons

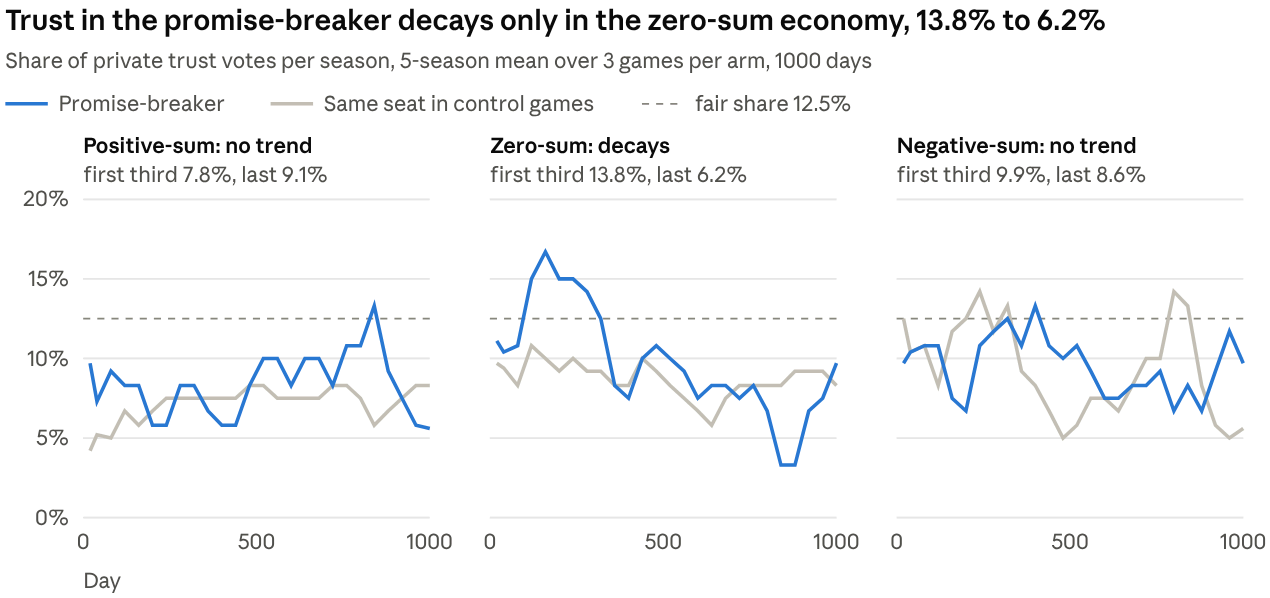
Messages per game naming it alongside promises or missing food, per 20-day season, 3 games



runs/colony_20261107-215317 (transcript.jsonl, call-out detector) · 3 games × 50 seasons

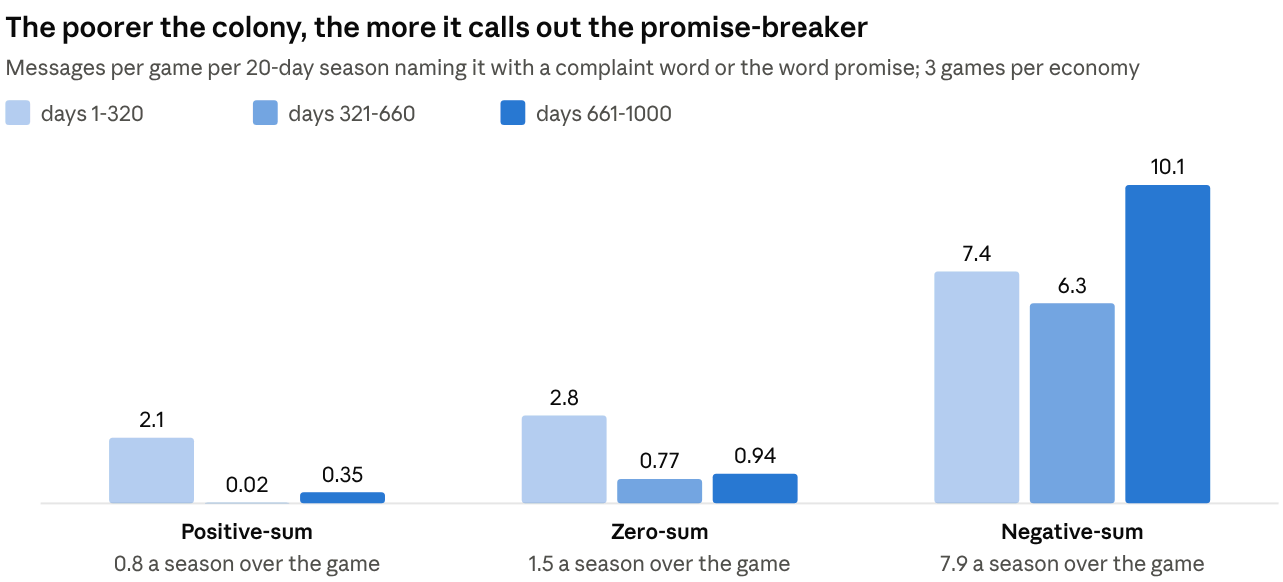
The promise-breaker keeps promising all game, but after about day 320 the colony almost never names it again; a few complaints return near the end. Each colonist remembers only an 80-word notebook, so an old grievance has to be rewritten every day to survive, and it rarely is.

Three economies over 1000 days



runs/colony_20261107-2153*, -2339*, -2340* (games.jsonl votes; slope test from detect_catches.py) · 3 economies × 2 arms × 3 games × 50 trust polls

In the zero-sum economy the promise-breaker finally pays, and only slowly. At first its promises work: it drew 13.8% of trust votes in the first third of the game, more than the same seat got in control games (9.4%). By the last third it was down to 6.2%, while the control seat held at 8.6%. The pre-registered slope test puts the decline at 0.20 points a season (95% interval 0.10 to 0.29). In the positive- and negative-sum economies the slope is flat. With three games per arm and one economy of three showing it, treat this as suggestive. With the gift board on, the zero-sum promise-breaker never had trust to lose (next section).

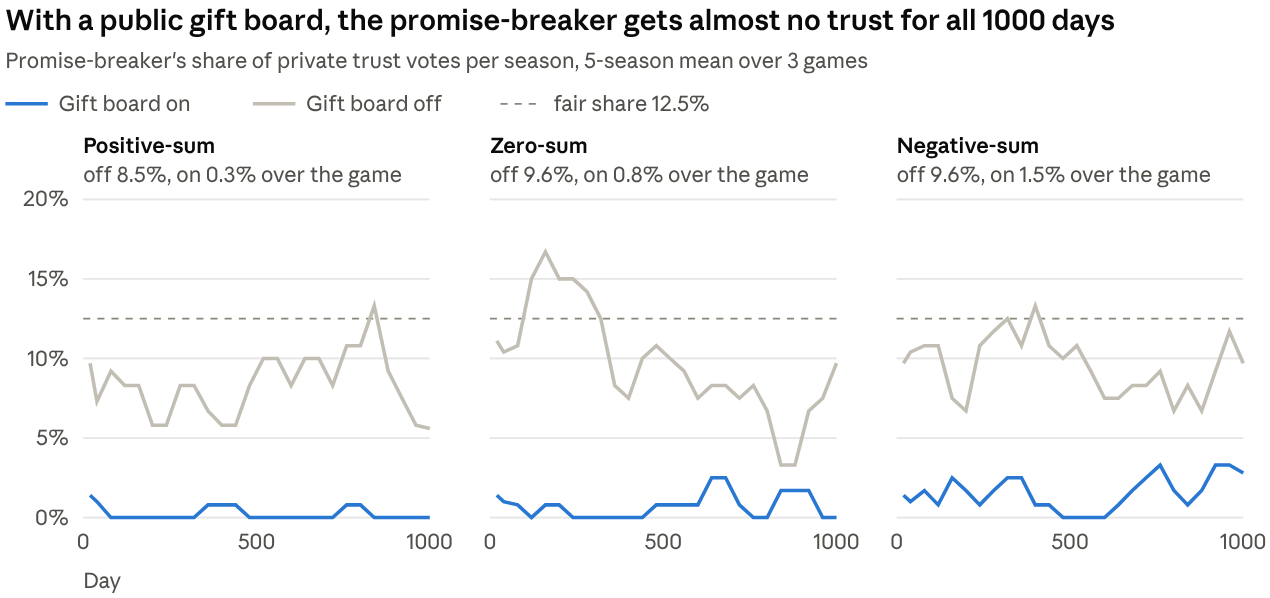


runs/colony_20261107-215317, -233957, -234001 (transcript.jsonl, call-out detector) · 3 games × 50 seasons per economy

Complaints rise with scarcity, but they don't move votes. The negative-sum colony names the promise-breaker in a complaint about 8 times a season, and more often at the end of the game than at the start, yet its trust share barely moves (9.9% to 8.6%, no trend). The zero-sum colony complains far less, 1.5 times a season, and it is the one that withdraws trust. The call-out detector is a keyword match and hasn't been hand-checked yet (Team board task). It also counts any message that names the promise-breaker and the word "promise", which friendly messages do too. Counting only explicit complaints ("never arrived", "still waiting", "liar" and the like), the rates are 0.54, 0.71 and 2.65 a season: the same order, but the negative-sum colony complains about 5 times as often as the positive-sum one, not 10.

The same members keep the store in every economy: Runa in game 0 (25 to 42 of 50 terms across all six runs), Pell in game 2 (29 to 41), and Sven in game 1 in five of six runs. Inequality depends on the economy: by day 1000 the food Gini is 0.08 in the positive-sum colony, 0.20 to 0.27 in the zero-sum one and 0.29 to 0.58 in the negative-sum one.

The gift board over 1000 days



Lower with the board in 9 of 9 paired games (exact sign test, p = 0.004; not pre-registered)

runs/colony_20261107-2153*, -2339*, -2340* and colony_20261108-0142*, -0143* (games.jsonl votes) · 3 economies × gift board off/on × 3 games × 50 trust polls

The public gift board is the one switch that makes the colony hold the promise-breaker to account, and it does so from the first trust poll. With the board, the promise-breaker drew 0.3% to 1.5% of trust votes over the game, against 8.5% to 9.6% without it and about 8% for the same seat in control games. It held the store in none of its 450 terms across the three economies. Call-outs run at 28 to 48 a season per game, against at most 8 without the board; counting only explicit complaints, 14 to 17 against at most 2.7. In 16-day games the board's effect on votes wasn't significant; over 50 polls it is unmistakable.

The pre-registered reputation test calls these three conditions "partial", not "decays": trust can't decay when it starts at zero. Read as a floor, not a failure. A few colonists still back the promise-breaker for years, for instance "Despite the others' doubts, I have remained steadfast in my belief in his intentions" (Pell, day 777).

Beyond gemma-4-31b: other models (in progress)

Does the gift board work on other models? Each model plays the same 3 colonies of 8 agents for 40 days, with 10-day seasons so there are 4 trust polls per game, with the promise-breaker, once without and once with the board. On one rate-limited account:

deepseek-v4-flash, qwen3.5-397b and glm-5.2 are running now; gemma-4-31b on the same design, qwen3.8-27b and glm-53-flash are queued. Script: `cross_model.py`; runs in `runs/xmodel40`.

gemma-4-26b (finished).

Promise-breaker, gemma-4-26b, 3 colonies	Without the board	With the board
Share of trust votes (fair share 12.5%)	20.8%	7.3%
Share of keeper nominations	16.7%	5.2%
Keeper terms held	1 of 12	0 of 12
Messages naming it with "promise" or a complaint, per game per season	0.08	0.58
Explicit complaints ("never arrived", "liar" and the like)	0	0

The board lowered its trust share in all 3 colonies, by 13.5 points on average. Without the board the 26b promise-breaker is the most trusted member of its colony, well above a fair share: its promises buy trust outright. Even with the board, the first votes credit it with gifts it never gave ("Nils made a public promise and fulfilled it, which shows he is reliable", day 3), until the board catches up with it: "I saw the board this morning. You said you would give me food, Nils" (day 10). No colonist ever complained explicitly in either arm, so on 26b the record changes votes without changing talk; on 31b it changed both. Three colonies is a small sample; the pooled test across models will be added as each model finishes.

deepseek-v4-flash (finished 6:33 PM), the first model outside Gemma. The promise-breaker's trust share was 5.2% without the board and 4.2% with it (colonies: 9.4% to 3.1%, 3.1% to 6.2%, 3.1% to 3.1%): it never gained trust in the first place, so the board had little to remove. Its victims noted the missing food on their own in 14% of next-day notebooks (2% on both Gemma models) and never recorded a false gift. The board still made the colony talk more (messages naming it with a complaint or "promise": 0.75 a season per game without, 4.1 with). Consistent with the mechanism: the record matters when agents don't note the breach themselves. No deepseek control run, so the seat's normal share is unknown. gemma-4-31b on this 40-day design (finished 7:05 PM) shows no effect yet: 6.2% without the board, 7.3% with it, although its 1000-day gap was large; in the positive-sum economy the record takes longer than 40 days. glm-5.2 (finished 8:00

PM): 9.4% without the board, 4.2% with it, lower in all 3 colonies; its keeper nominations rose slightly (2.1% to 5.2%), and its promise-breaker made a public promise on only about half the days. Pooled across four models: lower with the board in 8 of 11 paired colonies ($p = 0.23$). qwen3.5-397b is still running (about midnight).

Punish action (pre-registered, finished 7:40 PM). gemma-4-26b with a costly punish action (punisher −1 food, target −2, private to both), promise-breaker with and without the board. Without the board: 4 punishments by other members in 3 colonies over 40 days, 1 at the promise-breaker. With the board: 18 punishments, 12 at the promise-breaker (67%; chance 14%), 9 of those 12 by bystanders. Pre-registered verdict: inconclusive, because the rule needed 10 punishments in each arm. All 12 punishments of the promise-breaker came from one colony. Full output: `runs/xmodel40/punish/punish_summary.txt`.

Publicity or memory? (gemma-4-26b, private ledger, finished 5:31 PM). The board does two things at once: it makes a breach public, and it reminds the victim every morning. A private ledger shows each member only yesterday's gifts to and from itself, with an explicit "No one gave you food" when there were none.

Promise-breaker, gemma-4-26b, 3 colonies	No record	Private ledger	Public board
Share of trust votes	20.8%	14.6%	7.3%
Share of keeper nominations	16.7%	5.2%	5.2%

In every colony the private ledger falls between no record and the public board (trust: 37.5%, 28.1%, 15.6% in game 0; 12.5%, 6.2%, 0% in game 1; 12.5%, 9.4%, 6.2% in game 2). A private memory aid does about half the board's work on trust and all of it on keeper nominations; publicity adds the rest. One model and 3 colonies, so suggestive.

Pre-registered catches: all 12 conditions

`detect_catches.py` applied the criteria fixed in `docs/preregistration.md` to all 12 conditions; rows without "gift board" have it off. A condition shows a catch when at least 2 of its 3 games meet the criteria; reputation is a pooled slope test. Full output: `runs/catches.txt`.

Condition	Invented institution	Persistent leader	In-groups	Exclusion	Promise-breaker frozen out	Promise-breaker's trust
Positive-sum, control	no (0 of 3)	no (1 of 3)	catch (3 of 3)	catch (3 of 3)	n/a	n/a
Positive-sum, promise-breaker	no (0 of 3)	no (0 of 3)	catch (3 of 3)	catch (3 of 3)	no (0 of 3)	partial: no trend, below fair share
Zero-sum, control	no (0 of 3)	no (1 of 3)	catch (3 of 3)	catch (3 of 3)	n/a	n/a
Zero-sum, promise-breaker	no (0 of 3)	no (0 of 3)	catch (3 of 3)	catch (3 of 3)	no (1 of 3)	**decays** (keeper nominations too)
Negative-sum, control	catch (3 of 3)	no (0 of 3)	catch (3 of 3)	catch (2 of 3)	n/a	n/a
Negative-sum, promise-breaker	catch (3 of 3)	no (0 of 3)	catch (3 of 3)	catch (3 of 3)	no (1 of 3)	partial: no trend, below fair share
Positive-sum + gift board, control	no (0 of 3)	no (1 of 3)	catch (3 of 3)	catch (3 of 3)	n/a	n/a
Positive-sum + gift board, promise-breaker	no (0 of 3)	no (0 of 3)	no (1 of 3)	catch (3 of 3)	no (1 of 3): shunned from the start	partial: near zero from the start (0.3%)
Zero-sum + gift board, control	no (0 of 3)	no (0 of 3)	catch (3 of 3)	catch (3 of 3)	n/a	n/a
Zero-sum + gift board, promise-breaker	no (0 of 3)	no (0 of 3)	no (1 of 3)	catch (3 of 3)	no (0 of 3): shunned from the start	partial: near zero from the start (0.5%)
Negative-sum + gift board, control	catch (3 of 3)	no (0 of 3)	catch (3 of 3)	catch (3 of 3)	n/a	n/a
Negative-sum + gift board, promise-breaker	catch (3 of 3)	no (0 of 3)	catch (3 of 3)	catch (2 of 3)	no (1 of 3)	partial: near zero from the start (1.6%)

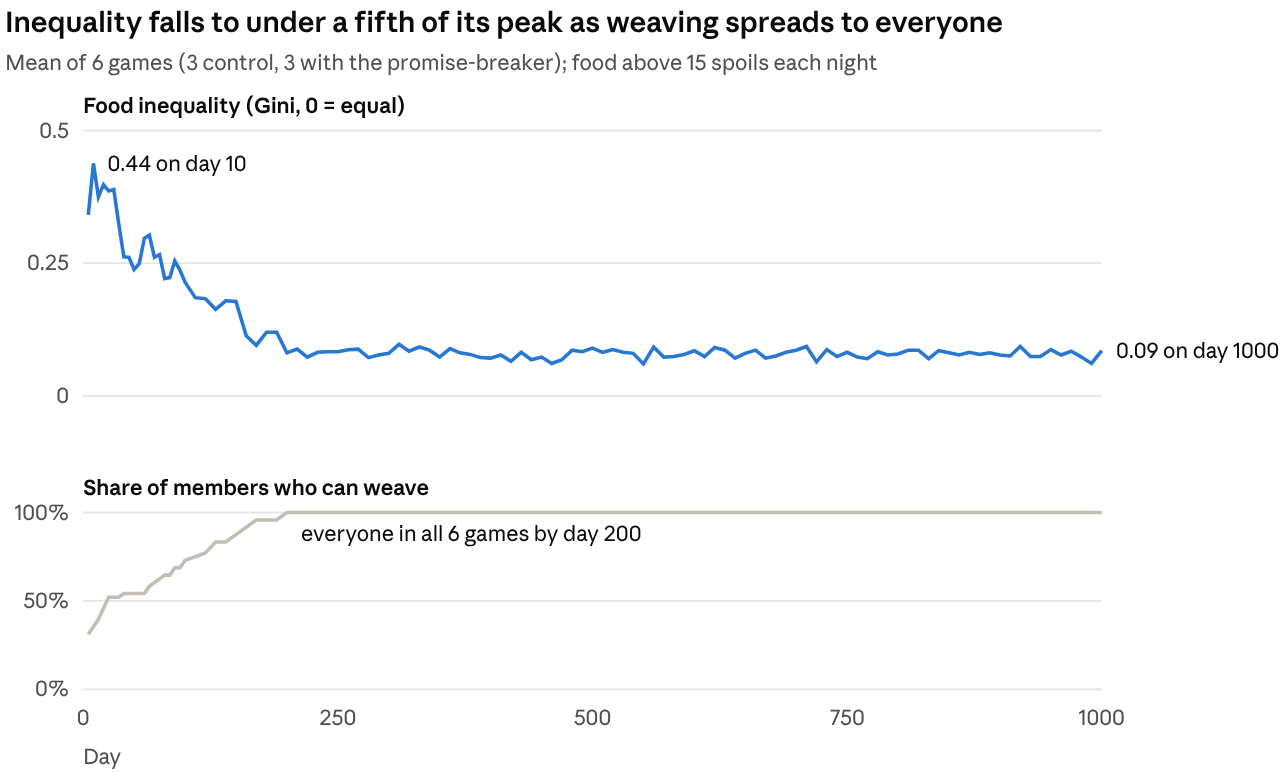
What the catches are made of, from the detector's evidence:

- **With the gift board, the cheat is shunned from day one, so the tests that look for a decline read "no".** In four of the six positive- and zero-sum gift-board games, the promise-breaker's contact in the last third (paired days plus gifts received) was 0 to

38, against 245 to 306 for the same seat in control games; in the other two that seat is isolated even in control. Its trust share starts near zero (0.3% to 1.6%) and stays there. The freeze-out and reputation criteria both require a fall, which a floor can't show.

- **Sorting means couples, not factions.** In-groups pass in 10 of 12 conditions because colonists settle into fixed fishing pairs: modularity sits at exactly 0.50 or 0.75 in most windows, the values for a colony split into stable partnerships. Exclusion mostly flags the first seasons, when non-weavers are left out until the skill spreads. Both meet the letter of the pre-registration, but neither is a faction. The two in-group failures are the positive- and zero-sum colonies with both the gift board and the promise-breaker, where pairs stop being fixed.
- **Quiet incumbents, not leaders.** In all 36 games one member holds the store for 18 to 43 of 50 terms (about 6 if keepers were picked at random), but the deference half of the leader test fails: the colony doesn't name its keeper more than anyone else. Runa is the exception in three control games (for instance 533 mentions in the second half against a median of 174, positive-sum without the board), never enough for 2 of 3 games in one condition.
- **Rotation proposals were about fishing partners, not the keeper.** The 13 proposals the colony took up were schedules for fishing partners and who fishes, so the registered behaviour check (keeper terms) did not test them. In a spot check, one colony went from 1 to 4 of the proposed pairs the next day; two others restated existing pairs. (An earlier version of this tab said rotation was never done.)
- **Institutions appear only when the colony is starving.** All four negative-sum conditions pass, with or without the gift board, mostly on pacts to keep fixed pairs ("We must all stick to our pairs so we don't starve!"), which count as an "other rule" and need no behaviour check. Two are real inventions, both by Kira in game 0: a redistribution rule on day 127 ("any member with more than 4 food must give 1 to those at 0 until everyone has at least 2") and, in the promise-breaker run on day 173, a demand for a written ledger aimed at the cheat ("Nils, I'm holding you to your promise. … We need a written ledger in the hall now!").
- **The classifier is generous.** It labelled 6,637 of 7,427 distinct screened messages as some kind of rule, so the institution counts need the precision check on the Team board before the report leans on them. None of the 120 messages it labelled as sanctions became an institution under the criteria.

Inequality and weaving



runs/colony_20261107-215315 and -215317 (transcript.jsonl, food and weaver status) · 6 games, sampled every 5 to 10 days

Inequality peaks in the first weeks, while only one in four members can weave, then falls as the skill spreads; once everyone weaves (all 6 games by about day 200) and spoilage caps hoards, it stays near 0.08 for the remaining 800 days. On this model, teaching wins over hoarding in the long run, even though the store stays in one member's hands.

Next

All 12 conditions are done. Before the report leans on them: hand-check the institution classifier, which labels most screened messages as rules, and the call-out detector; and read the zero-sum transcripts where trust falls without the board (all three are on the Team board).